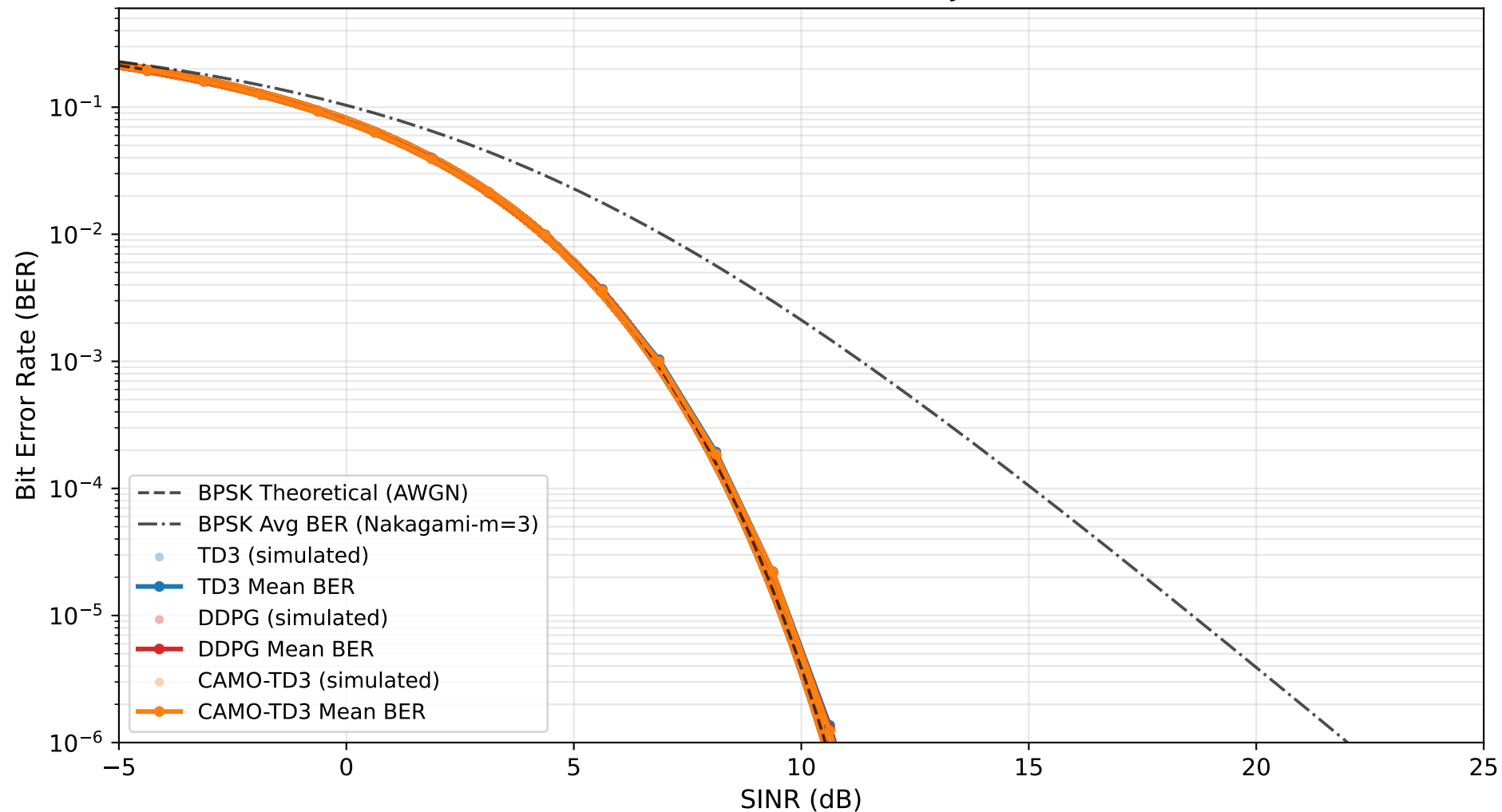


# CRN Power Control: TD3 vs DDPG vs CAMO-TD3 Nakagami-m Fading Channel Performance Report

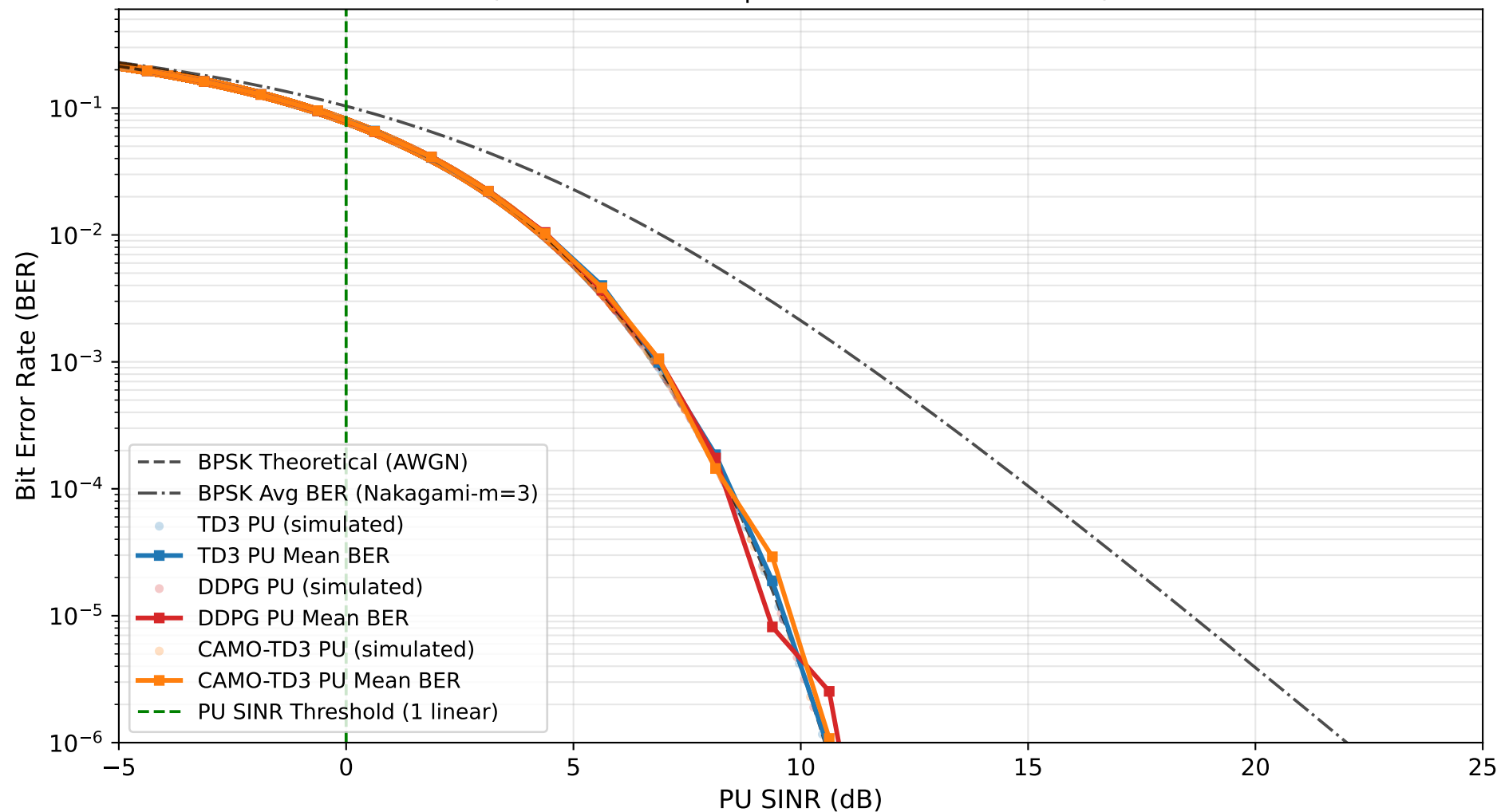
Nakagami-m = 3.0 | Episodes = 1000 | Steps/Episode = 200 | SINR Threshold = 1.0 dB

| Metric                    | TD3      | DDPG     | CAMO-TD3 | Winner   |
|---------------------------|----------|----------|----------|----------|
| Avg Reward (last 100 ep)  | 236.5460 | 245.1138 | 243.4966 | DDPG     |
| SU Throughput (bits/s/Hz) | 2.0160   | 2.0943   | 2.0877   | DDPG     |
| PU Throughput (bits/s/Hz) | 0.5572   | 0.5230   | 0.5224   | TD3      |
| Outage Probability        | 0.1235   | 0.1068   | 0.1041   | CAMO-TD3 |
| Average BER               | 0.030381 | 0.026838 | 0.026644 | CAMO-TD3 |
| Training Time (s)         | 4073.4   | 3838.4   | 8631.3   | -        |

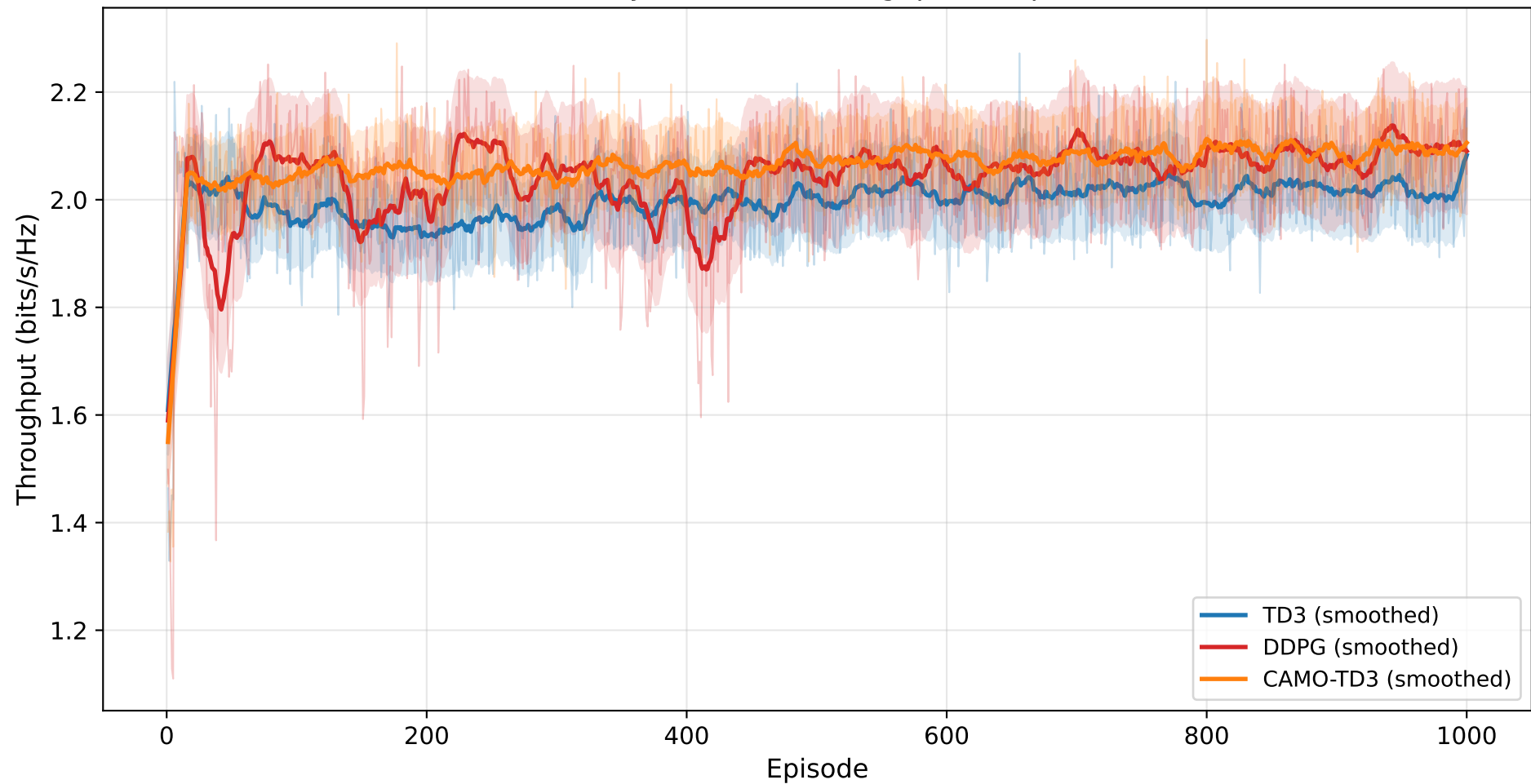
### SINR vs BER - Nakagami-m=3 Fading Channel (BPSK Modulation, Secondary User)



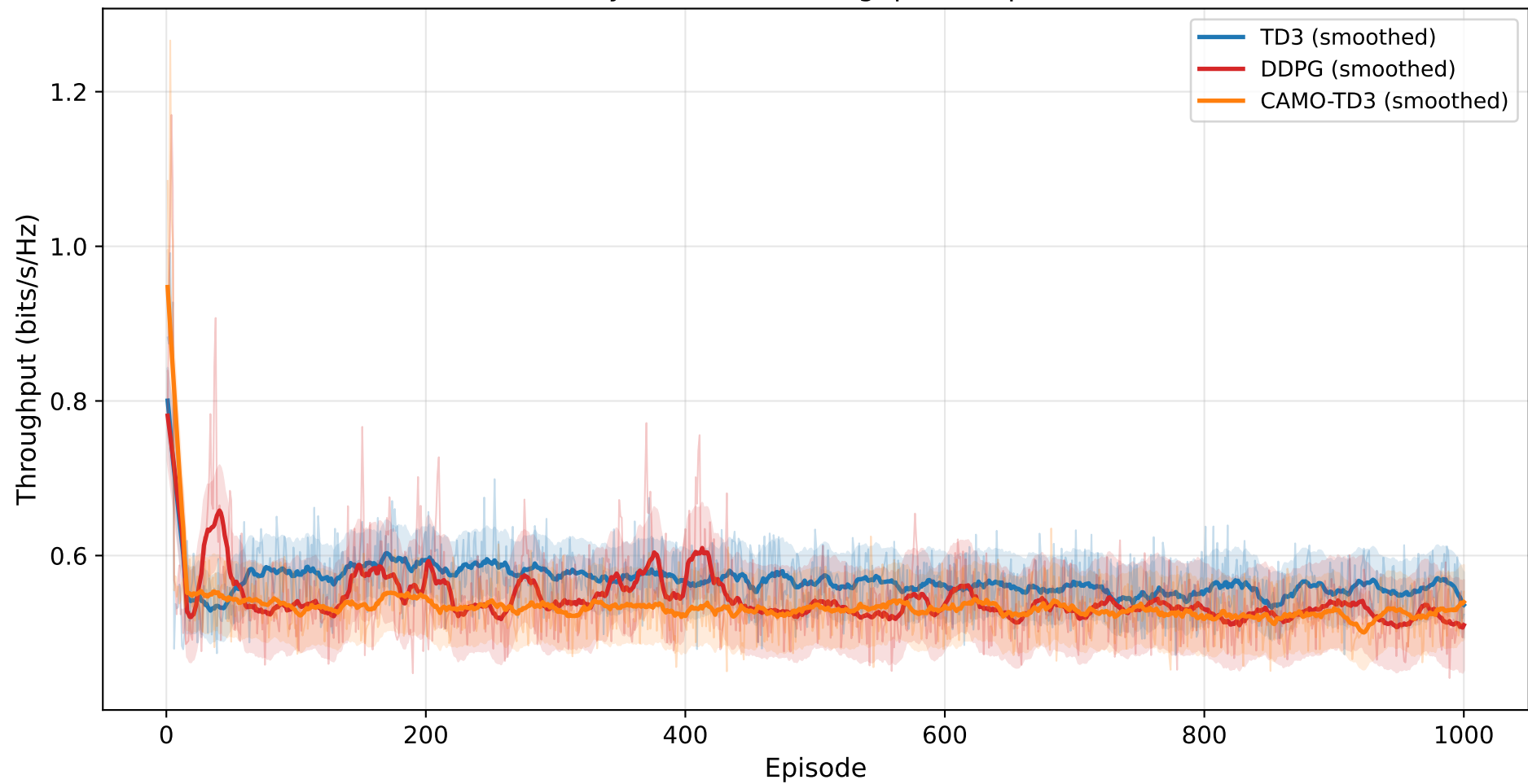
### Primary User SINR vs BER - Nakagami-m=3 Fading Channel (BPSK Modulation | Effect of SU Interference)



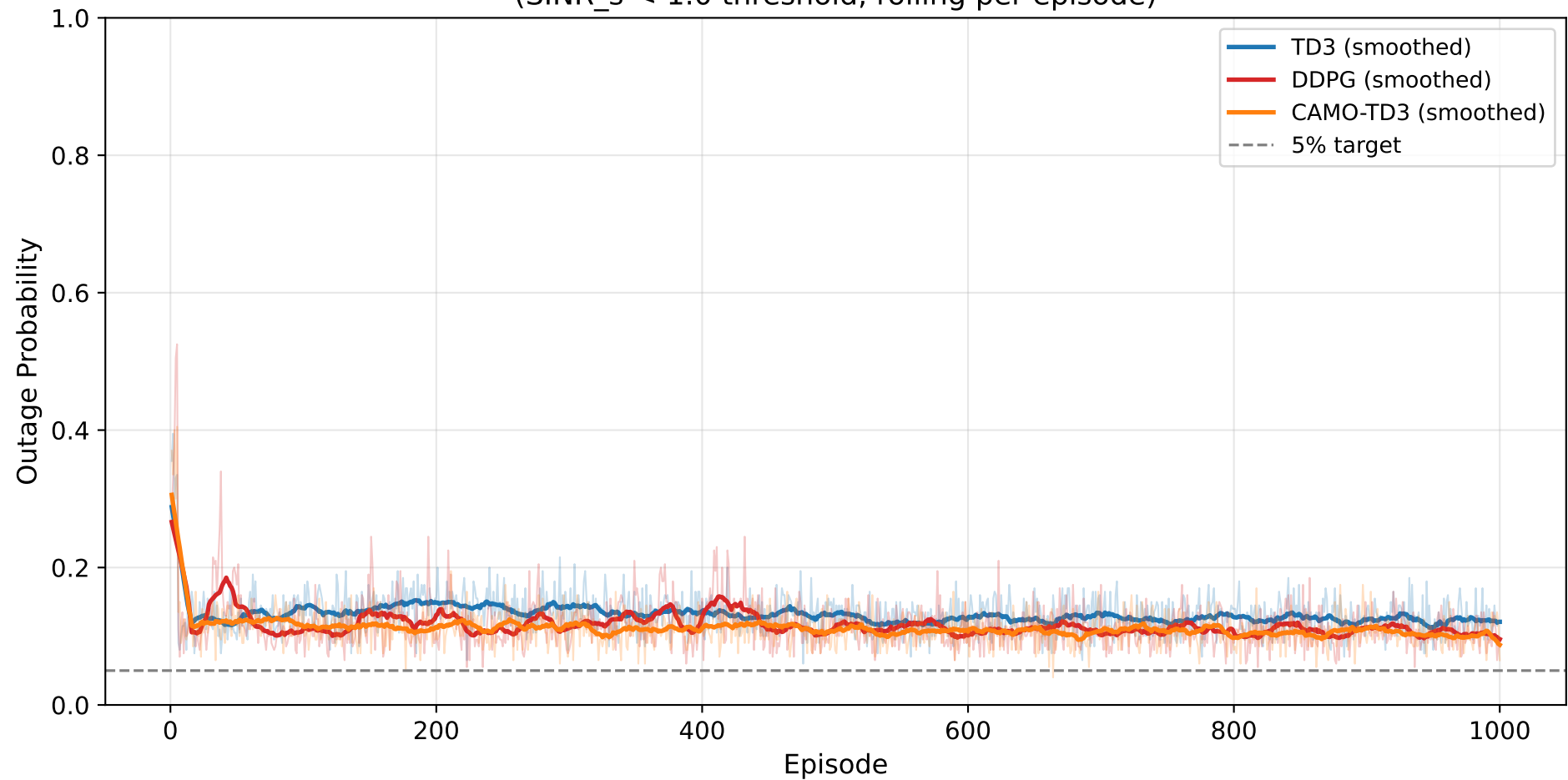
# Secondary User (SU) Throughput vs Episodes



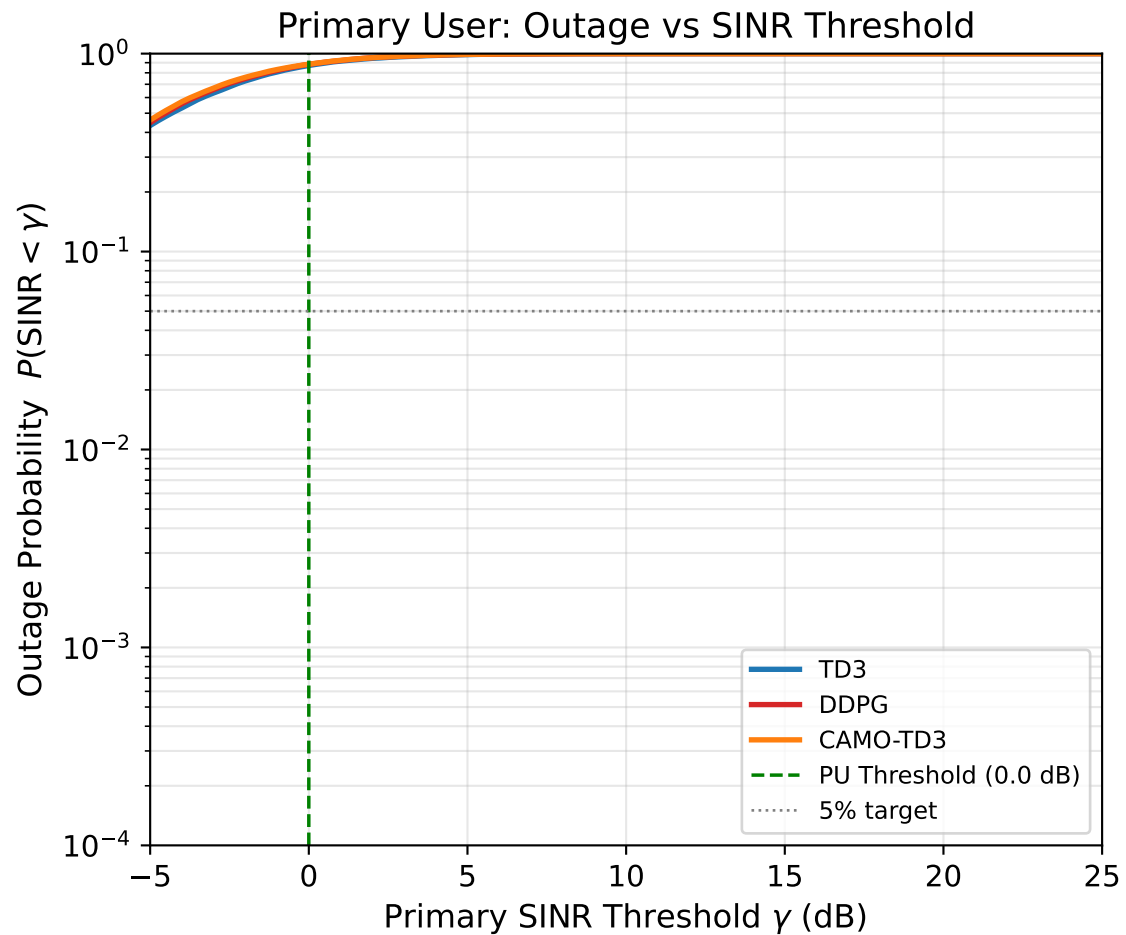
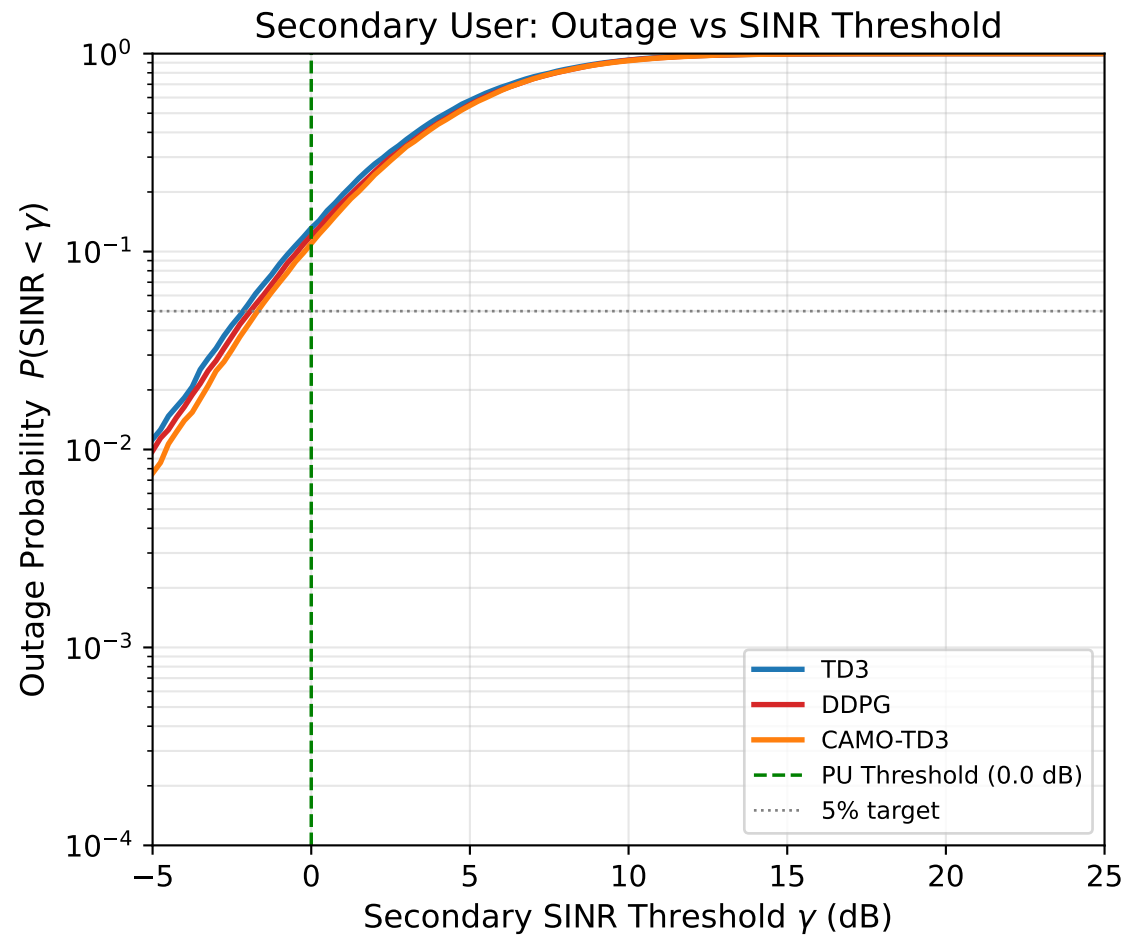
Primary User (PU) Throughput vs Episodes



Outage Probability vs Episodes  
(SINR<sub>s</sub> < 1.0 threshold, rolling per episode)

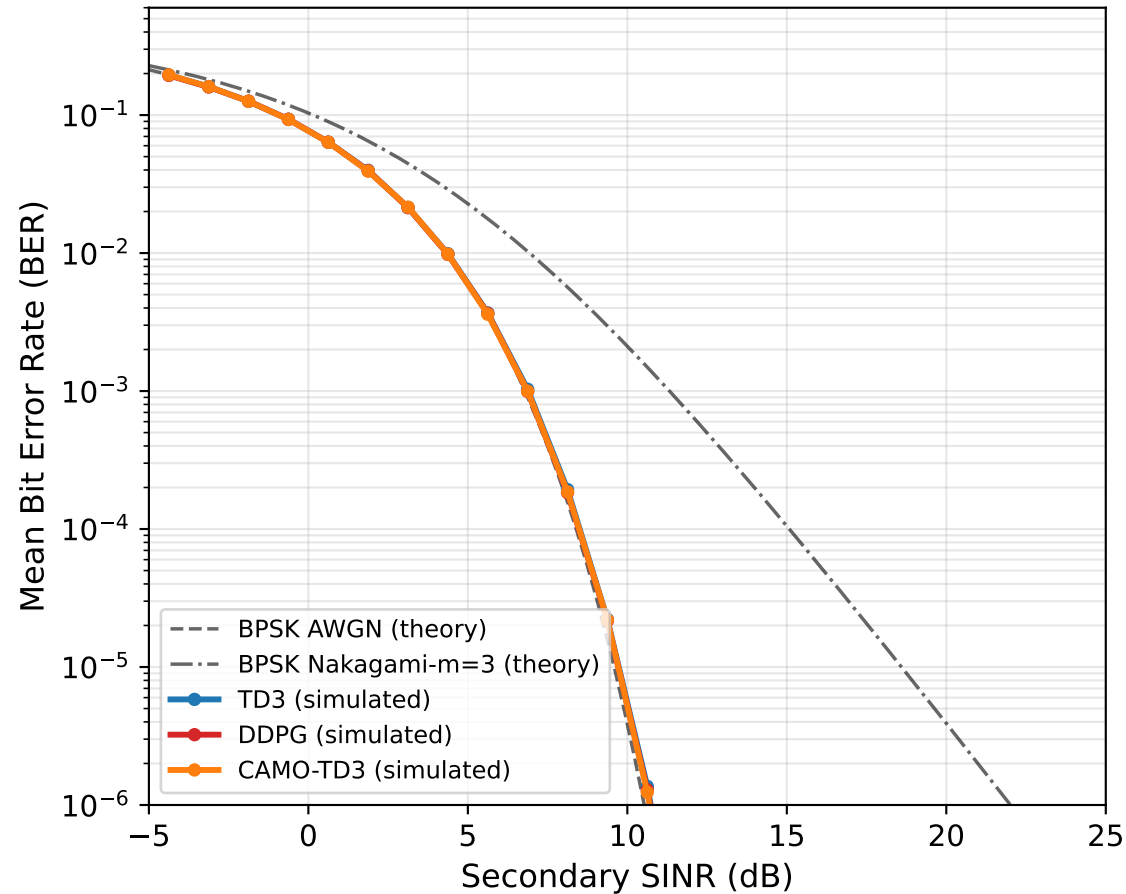


# Outage Probability vs SINR Threshold (Empirical CDF) — Imperfect CSI

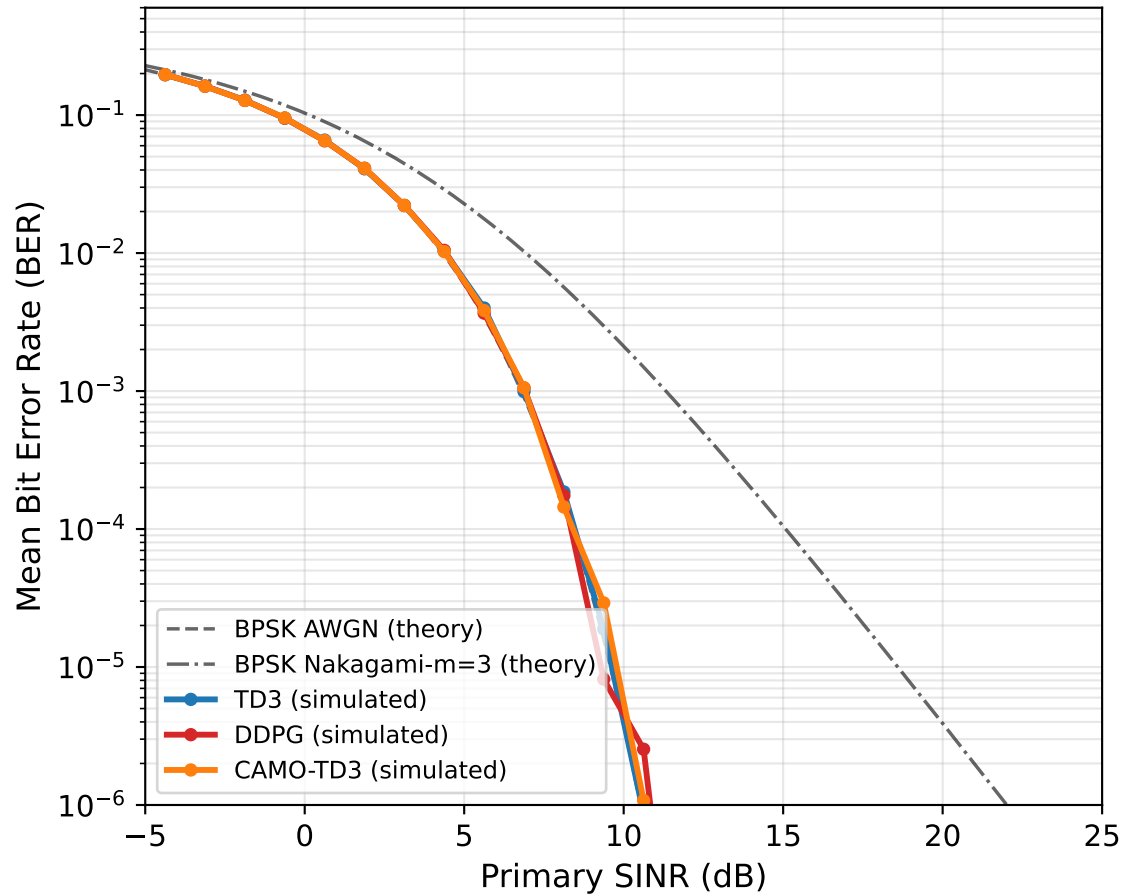


# BER vs SINR — Algorithm Comparison under Imperfect CSI

## Secondary User: BER vs SINR

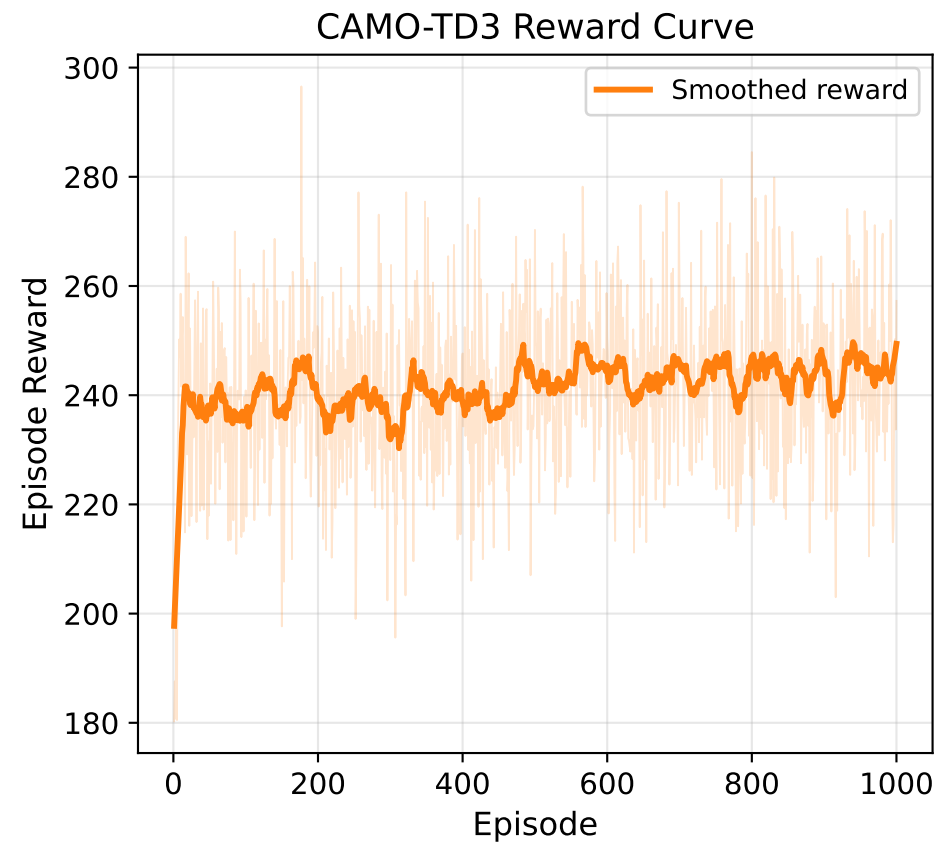
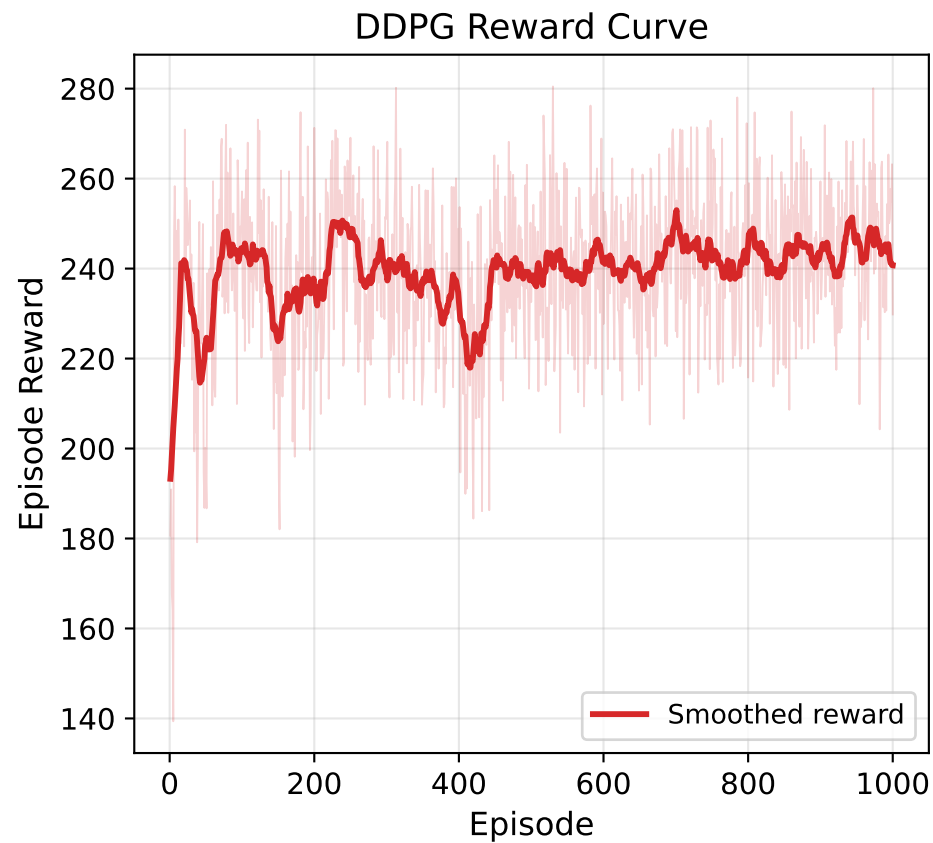
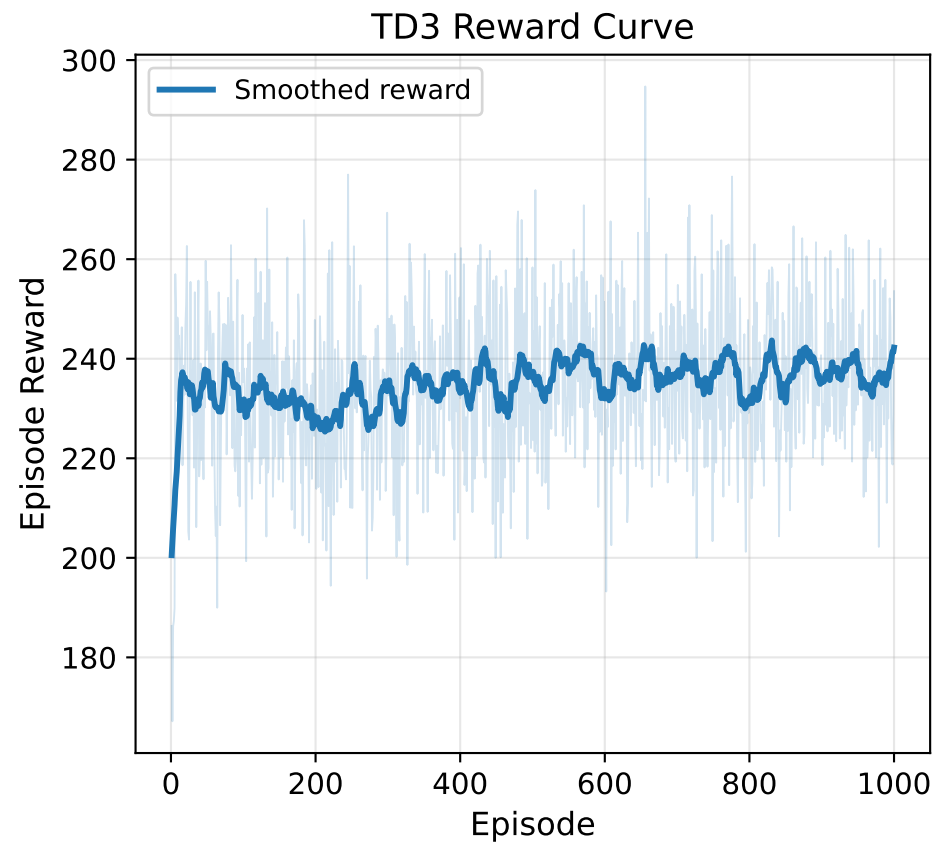


## Primary User: BER vs SINR





## Episode Reward Curves - TD3 vs DDPG vs CAMO-TD3



TD3 vs DDPG vs CAMO-TD3 - Reward Comparison

